A nonalgebraic Frobenius action
Counterexample to TLMC 00000000523

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Abstract

For the numerical semigroup $S = \mathbb{N} = \langle 1 \rangle$ over $\mathbb{F}_2$, the Frobenius action on first local cohomology has no nonzero annihilating polynomial over $\mathbb{F}_2$. Its minimal polynomial therefore does not exist, despite the semigroup having multiplicity one. This disproves the asserted equality of minimal-polynomial degree and semigroup multiplicity. The accompanying Lean development verifies the concrete principal-Čech quotient and the nonannihilation statement for every finite coefficient polynomial.

1 Statement and conventions

The original problem states that, for numerical semigroup rings, the degree of the minimal polynomial of Frobenius on $H^1_{\mathfrak{m}}(R)$ equals the multiplicity of the semigroup. It then asserts an additional formula of Wilf type. A counterexample to the first universally quantified assertion suffices; no interpretation or disproof of the further unspecified formula is needed.

A numerical semigroup is an additively closed subset of $\mathbb{N}$ containing zero with finite complement. Its multiplicity is its least positive element. Thus $S = \mathbb{N} = \langle 1 \rangle$ is allowed and has multiplicity $m(S) = 1$. The problem does not restrict to singular semigroups or require $m(S) > 1$.

Choose the prime field $k = \mathbb{F}_2$ and the semigroup ring

$$R = k[t^s : s \in S] = k[t], \qquad \mathfrak{m} = (t).$$

All operator polynomials below have coefficients in k. Frobenius is k-linear here because $c^2 = c$ for every $c \in k$. We do not treat Frobenius as R-linear, which would be false.

2 The actual local-cohomology module

For a principal ideal, the extended Čech complex is

$$0 \longrightarrow R \longrightarrow R_t \longrightarrow 0,$$

with R in degree zero and localization in degree one. The standard local-cohomology computation therefore gives

$$H^1_{(t)}(k[t]) \cong k[t, t^{-1}]/k[t]. \tag{1}$$

See Stacks Project, Lemma 47.9.1 and Lemma 51.2.1 [2, 3]. Here the ring is Noetherian, so the usual ideal-support and supported-cohomology definitions agree.

Every Laurent polynomial uniquely splits into a polynomial and a finite sum of negative powers. Consequently the classes

$$e_j = [t^{-j}], \qquad j = 1, 2, \ldots$$

form a k-basis of the quotient in (1). In particular, equality of quotient classes is exactly equality of all negative coefficients. This description is not a truncation: arbitrary finite negative supports are admitted.

The ring Frobenius sends a Laurent polynomial $a(t)$ to $a(t)^2$. In characteristic two all mixed terms cancel, giving

$$\left(\sum_i a_i t^i\right)^2 = \sum_i a_i t^{2i}.$$

It preserves $k[t]$ and hence induces the local-cohomology Frobenius F on the quotient. In particular,

$$F(e_j) = e_{2j}, \qquad F^r(e_1) = e_{2^r}. \tag{2}$$

Functoriality of the Čech construction identifies this induced map with Frobenius on local cohomology, rather than a separately invented endomorphism.

3 Counterexample

Theorem 1. *The endomorphism F of $H^1_{(t)}(\mathbb{F}_2[t])$ has no nonzero annihilating polynomial in $\mathbb{F}_2[X]$.*

Proof. Let $P(X) = \sum_{r=0}^{d} c_r X^r \neq 0$. By (2),

$$P(F)e_1 = \sum_{r=0}^{d} c_r e_{2^r}.$$

The positive integers 2^r are distinct. The displayed vectors are therefore distinct elements of the negative-power basis. Choosing any r with $c_r = 1$, the coefficient of t^{-2^r} in this quotient class is one. Thus $P(F)e_1 \neq 0$, so $P(F)$ is not the zero operator. □

In particular neither of the monic linear polynomials X and $X+1$ annihilates F: their values on e_1 are e_2 and $e_2 + e_1$, respectively. A minimal polynomial of degree $m(S) = 1$ cannot exist. Indeed no minimal polynomial exists at all. If a convention assigns degree $+\infty$ to a nonalgebraic endomorphism, that value is still not one.

Some authors use "numerical semigroup ring" for the completed local ring $k[[t^S]]$. The same mathematical counterexample persists under that convention: $k[[t]]_t = k((t))$, and $k((t))/k[[t]]$ has the identical finite negative-power basis and identical Frobenius action. The supplied Lean code implements the polynomial/Laurent-polynomial convention, not formal power series. Thus this paragraph is a mathematical extension, not an extra claimed Lean theorem.

4 Lean verification and precise coverage

The development uses Lean 4.31.0 with only its standard library. In `Main.lean`:

- `Laurent` is an actual finitely supported function $\mathbb{Z} \to \mathbb{F}_2$, with an explicit support bound. Boolean field identities are checked. `Polynomial` is the subspace whose negative coefficients vanish.
- `clear_denominators` proves that shifting any Laurent polynomial by a sufficiently large nonnegative power makes it a polynomial, and shifting back recovers it. This verifies the concrete denominator description R_t.
- `cechSetoid` identifies exactly the Laurent polynomials whose difference is polynomial. `H1` is Lean's genuine quotient by that equivalence relation, not an assigned dimension or table of values.

- The exponent-doubling map descends to the quotient and its additivity and $\mathbb{F}_2$-linearity are proved. The classes e_1, e_2 are proved nonzero and distinct.
- `polyEval` is Horner evaluation at this linear endomorphism. `polyEval_coefficient` proves that the negative coefficient at -2^r of $P(F)e_1$ equals the coefficient of X^r in P.
- `no_nonzero_annihilator` quantifies over every finite coefficient list with some nonzero coefficient, not merely a bounded search. `counterexample` also supplies the numerical-semigroup and multiplicity-one certificate and the failure of all monic degree-one annihilators.

The additional modules `Square.lean` and `Bridge.lean` supply an explicit algebraic bridge: multiplication is ordinary finite double-list convolution, its mixed square terms cancel over $\mathbb{F}_2$, and its square equals exponent doubling. Every bounded coefficient Laurent polynomial is proved to have a finite sparse representation. Thus the square comparison applies to the entire Laurent space, not just a chosen list of witnesses. The quotient-level map is proved to agree with those squares.

The general derived-functor/Čech comparison theorem is used mathematically through [2, 3]; it is not reimplemented as a general Lean local-cohomology library. The concrete quotient model and operator statement are formalized. Formalization coverage must not be conflated with formalization of all general commutative algebra underlying the standard model.

5 Reproducibility

The pinned toolchain is Lean 4.31.0, commit `68218e876d2a38b1985b8590fff244a83c321783`. The script `VERIFY.sh` cleans and builds the project, prints theorem axiom dependencies, and invokes `leanchecker` for independent kernel replay. No `sorry`, custom conclusion axioms, compiler-trust tactic, or kernel-check bypass is used. The proof is exact and symbolic; no empirical computation is being substituted for a quantified theorem. `build_pdf.sh` reproduces this document using `pdflatex`. Logs accompany the submission.

References

[1] The Last Math Competition, Problem 00000000523. Original blob `1350fd5092d75704112961ab5683d7b6601e5b55`.
https://github.com/The-Last-Math-Competition/The-Last-Math-Competition/blob/main/conjectures/00000000523.md.

[2] The Stacks Project, Section 47.9, Lemma 47.9.1 (extended Čech complex for local cohomology). Accessed 22 October 2026.
https://stacks.math.columbia.edu/tag/0952.

[3] The Stacks Project, Lemma 51.2.1 (comparison with supported cohomology). Accessed 22 October 2026.
https://stacks.math.columbia.edu/tag/0A6T.